

Cedric PIZINA

Statistics and Data Science | Climate Dynamics

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Profile

Graduate student in Statistics and Data Science at Clemson University with an interdisciplinary background in climate physics, hydraulic engineering, and mathematical sciences. Interested in climate-data analysis and numerical and statistical modelling. Recognized as a top-performing student and valued for adaptability, rapid learning, and an interdisciplinary perspective.

Education

Clemson University PhD Program in Statistics and Data Science	Aug. 2026 - Present Clemson, SC
International Centre for Theoretical Physics (ICTP) Pre-PhD Diploma in Physics of Climate and Earth System Physics GPA: 4.0	Sep. 2025 - Aug. 2026 Trieste, Italy
AIMS South Africa Stellenbosch University Structured Master's in Mathematical Sciences GPA: 4.0	Sep. 2024 - Jul. 2025 Cape Town, South Africa
École Supérieure Polytechnique d'Antananarivo University of Antananarivo Master of Engineering in Hydraulics and Development GPA: 3.85	Jan. 2022 - Aug. 2024 Antananarivo, Madagascar
École Supérieure Polytechnique d'Antananarivo University of Antananarivo Bachelor of Engineering in Hydraulics and Fluid Mechanics GPA: 3.85	Jul. 2018 - Aug. 2021 Antananarivo, Madagascar

Research Experience

Postgraduate Diploma Thesis Uncovering the failure of the early-winter IOD-NAO teleconnection in SEAS5 and its sensitivity to different atmospheric mean states - Analyzed climate-model output using partial regression to identify the causes of teleconnection failure. - Conducted mean-state sensitivity experiments using a dry Linear Baroclinic Model.	Sep. 2025 - Aug. 2026
MSc Project Monitoring Southern African droughts under El Niño-Southern Oscillation (ENSO) and Indian Ocean Dipole (IOD) impacts - Analyzed the joint impact of ENSO and IOD phases on Southern Africa's climate during 1980-2023. - Assessed future trends in precipitation, temperature, and soil moisture through 2100 using CMIP6 models. - Forecasted future mean trends in the Southern African Drought Index using machine learning.	Mar. 2025 - Jun. 2025
MEng Project Diagnostic of the drinking-water network of Sainte Marie, Madagascar - Led site data collection and diagnostic analysis to identify operational problems and their causes. - Developed an optimized EPANET model for system diagnosis and optimization. - Proposed improvements to network efficiency.	Jul. 2023 - May 2024

Teaching and Professional Experience

Clemson University

Lab Instructor, Introduction to Statistics (STAT 2300)

- Conduct weekly laboratory sessions using JMP and grade laboratory assignments.

Aug. 2026 - Present

Clemson, SC

GRET Madagascar

Junior Hydraulic Engineer

- Contributed to the design of drinking-water systems and watershed protection plans for Ambatofotsy and Vinaninony.
- Conducted flow, pressure, and water-quality measurements at production sites in Andramasina and Sainte Marie.
- Contributed to field data collection by developing surveys and coordinating on-site quality controls.

Jul. 2023 - May 2024

Antananarivo, Madagascar

Selected Coursework

Geophysical Fluid Dynamics · Atmospheric Physics · Numerical Methods · Earth System Modeling · Ocean Dynamics · Biogeochemical Cycles · Applied Statistical Models · Statistical Inference

Awards and Leadership

Mastercard Foundation Scholarship | AIMS South Africa

Mastercard Foundation Scholar Network Member

Awarded a fully funded scholarship to complete a master's program at AIMS and develop leadership skills through community service.

Sep. 2024 - Jul. 2025

Academic Class Prize

Top achiever in undergraduate and graduate studies

Recognized as a top achiever at the École Supérieure Polytechnique d'Antananarivo and received an academic excellence award at AIMS South Africa.

Jul. 2018 - Jul. 2025

Community Engagement and Workshops

Muizenberg Library | AIMS South Africa

Mathematics and Physics Tutor

Tutored middle- and high-school students in mathematics and physics.

Jan. 2025 - Jul. 2025

Cape Town, South Africa

AIMS House of Science | African Gong

Africa Scientifique 2025

Attended a three-day workshop on scientific communication, discovery, and capacity building.

Apr. 2025

Cape Town, South Africa

University of Cape Town | AIMS South Africa

Design Thinking

Completed a two-day workshop focused on design thinking and teamwork skills.

Sep. 2024

Cape Town, South Africa

Technical Skills

Programming

Python, Fortran, SageMath, R, LaTeX

Engineering and GIS

AutoCAD, EPANET, QGIS, ArcGIS, HEC-HMS, HEC-RAS

Office

Microsoft Word, Excel, PowerPoint, and Outlook

Languages

Malagasy (native), French (C1), English (C1)

References

Dr. Fred Kucharski

International Centre for Theoretical Physics, Italy

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Dr. Riccardo Farneti

International Centre for Theoretical Physics, Italy

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